MATH 202 Your name: _____

Learning target A3, version 1

Use definite integrals to compute the volume of a solid of revolution. Decide whether to use washers or discs.

Draw the region bounded by $y = 3\sqrt{x}$, the x-axis, and the line $x = 9$.

(a) Suppose this region is revolved about the x-axis.

- Is a representative slice going to be horizontal or vertical?
- Should we use disks or washers?
- What is the approximate volume of a representative slice?
- Write down a definite integral that computes the volume of this solid of revolution.

(b) Now, suppose this region is revolved about the line $x = -4$.

- Is a representative slice going to be horizontal or vertical?
- Should we use disks or washers?
- What is the approximate volume of a representative slice?
- Write down a definite integral that computes the volume of this solid of revolution.

MATH 202 Your name: _____

Learning target A3, version 2

Use definite integrals to compute the volume of a solid of revolution. Decide whether to use washers or discs.

Draw the finite region bounded by $y = \frac{1}{2}(x-3)^2 + 2$ and $x = 5 - y$.

(a) Suppose this region is revolved about the y-axis.

- Is a representative slice going to be horizontal or vertical?
- Should we use disks or washers?
- What is the approximate volume of a representative slice?
- Write down a definite integral that computes the volume of this solid of revolution.

(b) Now, suppose this region is revolved about the line $y = 1$.

- Is a representative slice going to be horizontal or vertical?
- Should we use disks or washers?
- What is the approximate volume of a representative slice?
- Write down a definite integral that computes the volume of this solid of revolution.